

Derivada Covariante:

→ Como comparar vetores em pontos diferentes.

$$\frac{\partial V}{\partial q^i} = \frac{\partial (V^k e_k)}{\partial q^i} = \frac{\partial V^k}{\partial q^i} e_k + V^k \frac{\partial e_k}{\partial q^i}$$

as bases mudam.

$$\frac{\partial e_k}{\partial q^i} = \Gamma^j_{ki} e_j$$

Assim

$$\frac{\partial V}{\partial q^i} = \frac{\partial (V^k e_k)}{\partial q^i} = \frac{\partial V^j}{\partial q^i} e_j + V^k \Gamma^j_{ki} e_j = \left(\frac{\partial V^j}{\partial q^i} + V^k \Gamma^j_{ki} \right) e_j$$

$$\left(\frac{\partial V}{\partial q^i} \right)^j \equiv V^j{}_{;i}$$

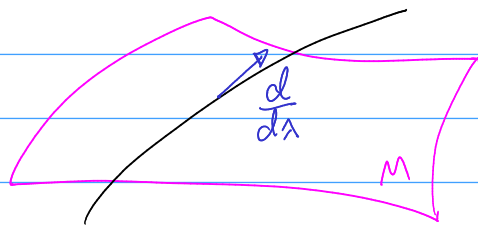
$$\frac{\partial V}{\partial q^i} = V^k{}_{;i} e_k \quad \leadsto \quad \nabla_{e_i} V = \nabla_i V$$

→ bases q e ∂ são bases coordenadas

Derivada covariante e Geodésicas:

$$\frac{\partial V}{\partial q^i} = \left(\frac{\partial V^k}{\partial q^i} + V^j \Gamma^k_{ji} \right) e_k \quad \text{e} \quad e_k = \frac{\partial}{\partial q^k}$$

$$\frac{\partial V}{\partial q^i} = V^k{}_{;i} e_k$$



$$V = \frac{d}{d\lambda}$$

Curva integral de um campo vetorial em um segmento se no segmento o C.V é tangente a curva

E a curva integral é uma geodésica se $\frac{dV}{d\lambda} = 0$

$$V = V^i \frac{\partial}{\partial q^i} = \frac{d}{d\lambda} = \frac{dV^i}{d\lambda} \frac{\partial}{\partial q^i}$$

$$\text{e } \frac{\partial V}{\partial \lambda} = \frac{\partial V}{\partial q^i} \frac{\partial q^i}{\partial \lambda} = \frac{\partial V}{\partial q^i} V^i =$$

mas:
$$\frac{\partial V}{\partial q^i} = \left(\frac{\partial V^k}{\partial q^i} + V^j \Gamma^k_{ji} \right) e_k$$

Assim:

$$\frac{\partial V}{\partial \lambda} = \left(\frac{\partial V^k}{\partial q^i} + V^j \Gamma^k_{ji} \right) e_k V^i = V^k_{,i} V^i e_k = 0$$

p/ ser geodésica ✓

Portanto as componentes devem ser 0.

$$V^i \frac{\partial V^k}{\partial q^i}$$

1º termo

$$+ V^i V^j \Gamma^k_{ji} = 0$$

2º termo

1º termo: $\frac{\partial v^k}{\partial q^i} v^i = \frac{\partial v^k}{\partial q^i} \frac{\partial q^i}{d\lambda}$ e $\frac{d}{d\lambda} = \frac{dq^i}{d\lambda} \frac{\partial}{\partial q^i}$

então $\frac{d}{d\lambda} v^k = \frac{dq^i}{d\lambda} \frac{\partial v^k}{\partial q^i} = \frac{\partial v^k}{\partial q^i} v^i$

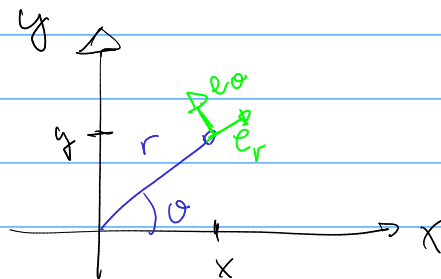
ainda $\frac{d}{d\lambda} \left(\frac{dq^k}{d\lambda} \right) = \frac{d^2 q^k}{d\lambda^2} = \frac{\partial v^k}{\partial q^i} v^i$

Temos:

$$\frac{d^2 q^k}{d\lambda^2} + \Gamma^k_{ij} \frac{dq^i}{d\lambda} \frac{dq^j}{d\lambda} = 0$$

↳ Símbolos de Christoffel

Exemplo: Derivada covariante em coord. polares



$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$\vec{r} = (r \cos \theta, r \sin \theta)$$

$$\vec{e}_r = \frac{\partial \vec{r}}{\partial r} = (\cos \theta, \sin \theta) \quad \text{e} \quad \vec{e}_\theta = (-r \sin \theta, r \cos \theta)$$

$$\begin{aligned} \vec{e}_r &= \cos \theta \vec{e}_x + \sin \theta \vec{e}_y \\ \vec{e}_\theta &= -r \sin \theta \vec{e}_x + r \cos \theta \vec{e}_y \end{aligned}$$

$$\vec{e}_x = \frac{\partial}{\partial x} \quad | \quad \vec{e}_y = \frac{\partial}{\partial y} \quad | \quad \vec{e}_r = \frac{\partial}{\partial r} \dots$$

$$V = V^r e_r + V^\theta e_\theta$$

$$\frac{\partial V}{\partial r} = \frac{\partial}{\partial r} (V^r e_r + V^\theta e_\theta) = \frac{\partial V^r}{\partial r} e_r + V^r \frac{\partial e_r}{\partial r} + \frac{\partial V^\theta}{\partial r} e_\theta + V^\theta \frac{\partial e_\theta}{\partial r}$$

Como $\frac{\partial e_r}{\partial r} = 0$ e $\frac{\partial e_\theta}{\partial r} = -\sin\theta e_x + \cos\theta e_y = \frac{e_\phi}{r}$

$$\frac{\partial V}{\partial r} = \frac{\partial V^r}{\partial r} e_r + \left(\frac{\partial V^\theta}{\partial r} + \frac{V^\theta}{r} \right) e_\theta$$

(similar) $\rightarrow$ fazer

$$\frac{\partial V}{\partial \theta} = \left(\frac{\partial V^r}{\partial \theta} - V^\theta r \right) e_r + \left(\frac{V^r}{r} + \frac{\partial V^\theta}{\partial \theta} \right) e_\theta$$

Vimos que

$$\frac{\partial V}{\partial q^i} = \left(\frac{\partial V^k}{\partial q^i} + V^j \Gamma^k_{ji} \right) e_k$$

$$\frac{\partial V}{\partial r} = \frac{\partial V^r}{\partial r} e_r + \left(\frac{\partial V^\theta}{\partial r} + \frac{V^\theta}{r} \right) e_\theta$$

inspecionando

$$\Gamma^\theta_{r\theta} = \frac{1}{r}$$

Derivada Covariante da 1-forma:

$$\sigma = \sigma_i e^i \quad e^i = dq^i \quad \text{e} \quad \langle e^i, e_j \rangle = \delta^i_j$$

$$\frac{d}{dq^i} (\overset{\in \mathbb{R}}{e^j} \cdot e_k) = 0 \quad \rightarrow \quad \frac{\partial e^j}{\partial q^i} e_k + e^j \frac{\partial e_k}{\partial q^i} = 0$$

generalizando p/ rel. de dualidade:

$$\frac{\partial}{\partial q^i} \langle e^j, e_k \rangle = \left\langle \frac{\partial e^j}{\partial q^i}, e_k \right\rangle + \left\langle e^j, \frac{\partial e_k}{\partial q^i} \right\rangle$$

$$\frac{\partial}{\partial q^i} \langle e^i, e_k \rangle = \left\langle \frac{\partial e^i}{\partial q^i}, e_k \right\rangle + \left\langle e^i, \frac{\partial e_k}{\partial q^i} \right\rangle = 0$$

$$\left\langle \frac{\partial e^i}{\partial q^i}, e_k \right\rangle = - \left\langle e^i, \frac{\partial e_k}{\partial q^i} \right\rangle \quad \text{with a green arrow pointing from } \frac{\partial e_k}{\partial q^i} \text{ to } \Gamma^n_{kj} e_n$$

Logo:

$$\left\langle \frac{\partial e^i}{\partial q^i}, e_k \right\rangle = - \langle e^i, e_n \rangle \Gamma^n_{kj}$$

$$\left\langle \frac{\partial e^i}{\partial q^i}, e_k \right\rangle = - \int^n \Gamma^n_{kj}$$

$$\left\langle \frac{\partial e^i}{\partial q^i}, e_k \right\rangle = - \Gamma^i_{kj}$$

Portanto

$$\Gamma^i_{kj} = - \left\langle \frac{\partial e^i}{\partial q^i}, e_k \right\rangle$$

$$\Gamma^i_{kj} = - \left\langle \Gamma^i_{nj} e^n, e_k \right\rangle$$

$$\Gamma^i_{kj} = \Gamma^i_{nj} \int^n = \Gamma^i_{kj}$$

Così:

$$\frac{\partial e^i}{\partial q^i} = - \Gamma^i_{kj} e^k$$

Seja $\sigma = \sigma_i e^i$

$$\frac{\partial \sigma}{\partial q^i} = \frac{\partial}{\partial q^i} (\sigma_j e^j) = \frac{\partial \sigma_j}{\partial q^i} e^j + \sigma_j \frac{\partial e^j}{\partial q^i}$$

$$\frac{\partial \sigma}{\partial q^i} = \frac{\partial \sigma_j}{\partial q^i} e^j - \sigma_j \Gamma^j_{mi} e^m$$

$$\begin{aligned} \frac{\partial \sigma}{\partial q^i} &= \sigma_{j,i} e^j - \sigma_j \Gamma^j_{mi} e^m = (\sigma_{m,i} - \sigma_i \Gamma^i_{mj}) e^m \\ &= \sigma_{m,j} e^m \end{aligned}$$

Divergente em coordo polares:

Seja $\vec{v} = (v^1, v^2, v^3)$

$$\vec{\nabla} \cdot \vec{v} = \left(\frac{\partial}{\partial x} \hat{e}_1 + \frac{\partial}{\partial y} \hat{e}_2 + \frac{\partial}{\partial z} \hat{e}_3 \right) (v^1 \hat{e}_1 + v^2 \hat{e}_2 + v^3 \hat{e}_3)$$

$$= \frac{\partial v^1}{\partial x} + \frac{\partial v^2}{\partial y} + \frac{\partial v^3}{\partial z}$$

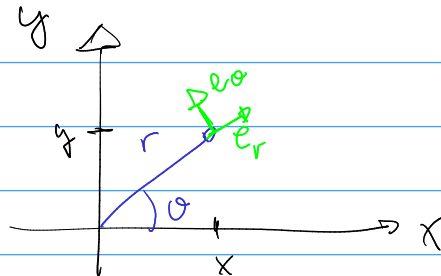
$$\begin{array}{l|l} x = q^1 & \vec{e}_1 = \hat{e}_1 \\ y = q^2 & \vec{e}_2 = \hat{e}_2 \end{array}$$

$$\vec{\nabla} \cdot \vec{v} = \left(\frac{\partial}{\partial q^1} \vec{e}_1 + \frac{\partial}{\partial q^2} \vec{e}_2 \right) (v^1 \vec{e}_1 + v^2 \vec{e}_2)$$

$$= \frac{\partial v^1}{\partial q^1} = v^1_{,1}$$

Para generalizar o divergente numa base qualquer, é preciso usar a derivada covariante:

$$V^i_{;i} \text{ ; } i = ?$$



$$x = r \cos \theta$$

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$$\begin{aligned} \vec{e}_r &= \cos \theta \vec{e}_x + \sin \theta \vec{e}_y \\ \vec{e}_\theta &= -r \sin \theta \vec{e}_x + r \cos \theta \vec{e}_y \end{aligned} \quad \left| \quad \vec{e}_x = \frac{\partial}{\partial x} \quad / \quad \vec{e}_y = \frac{\partial}{\partial y} \quad / \quad \vec{e}_r = \frac{\partial}{\partial r} \dots \right.$$

$$V = V^r \vec{e}_r + V^\theta \vec{e}_\theta$$

$$= V^r \partial_r + V^\theta \partial_\theta$$

$$\frac{\partial \vec{V}}{\partial q^i} = \left(\frac{\partial V^k}{\partial q^i} + V^j \Gamma^k_{ji} \right) \vec{e}_k \quad \text{onde} \quad \frac{\partial \vec{e}_i}{\partial q^j} = \Gamma^k_{ji} \vec{e}_k$$

contra notação

$$V^k_{;i} = V^k_{,i} + V^j \Gamma^k_{ji}$$

$$\text{Divergente de } V \left(\frac{\partial V^i}{\partial q^i} \rightsquigarrow V^i_{;i} \right)$$

Divergente de V

$$V^k{}_{;k} = V^k{}_{,k} + V^j \Gamma^k{}_{jk}$$

$$q^1 = r; \quad q^2 = \vartheta; \quad ; v^1 = v^r \text{ e } v^2 = v^\vartheta$$

$$V^k{}_{;k} = V^1{}_{;1} + V^2{}_{;2} = V^1{}_{,1} + V^2{}_{,2} + V^j \Gamma^1{}_{j1} + V^j \Gamma^2{}_{j2}$$

$$= V^1{}_{,1} + V^2{}_{,2} + V^1 \Gamma^1{}_{,1} + V^2 \Gamma^1{}_{2,1} + V^1 \Gamma^2{}_{12} + V^2 \Gamma^2{}_{22}$$

$$= V^r{}_{,r} + V^\vartheta{}_{,\vartheta} + \cancel{V^r \Gamma^r{}_{rr}} + \cancel{V^\vartheta \Gamma^\vartheta{}_{\vartheta r}} + \underbrace{V^r \Gamma^\vartheta{}_{r\vartheta}}_{\frac{1}{r}} + \cancel{V^\vartheta \Gamma^\vartheta{}_{\vartheta\vartheta}}$$

$$V^k{}_{;k} = \frac{\partial v^r}{\partial r} + \frac{\partial v^\vartheta}{\partial \vartheta} + \frac{v^r}{r} = \frac{1}{r} \frac{\partial}{\partial r} (r v^r) + \frac{\partial v^\vartheta}{\partial \vartheta}$$